

Side-Channel Analysis and Countermeasure Design on ARM-Based Quantum-Resistant SIKE

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Abstract—The implementations of post-quantum cryptographic algorithms have been newly explored, whereas, the protection against side-channel attacks shall be considered upfront, since it can have a non-negligible impact on security and performance. In this article, the security of supersingular isogeny key encapsulation (SIKE), a second-round candidate of NIST's on-going post-quantum standardization process, is thoroughly evaluated under side-channel analysis. First, the vulnerabilities of reference and optimized implementations of SIKE are thoroughly analyzed in terms of both horizontal and vertical side-channel leakage. After the optimized SIKE, which is based on Three-point Montgomery Differential Ladder algorithm, is proved to be constant-time and there is no horizontal leakage, a vertical vulnerability is analyzed based on the source code at the algorithmic level, and a theoretical differential power analysis (DPA) attack is proposed. In order to exploit this vulnerability, the differential electromagnetic attack (DEMA) is put into practice to extract the private key of SIKE based on a 32-bit ARM platform. To the best of our knowledge, this is the first practical side-channel attack at SIKE implemented on real ARM-based devices. Our experiments show that the DEMA needs only hundreds of electromagnetic traces to carry out the attack. More importantly, an efficient window-based countermeasure is proposed to eliminate the vertical leakage and prevent side-channel attacks with only a little overhead. The security of our countermeasure is carefully evaluated against most of well-known power analysis attacks. Through careful evaluation and comparison with other countermeasures, this method can lead to higher security at a very small cost in terms of time and memory.

Index Terms—PQC, SIKE, vulnerability detection, DPA, DEMA, window-based countermeasure

1 INTRODUCTION

SINCE the Diffie-Hellman (DH) key exchange scheme was first proposed in 1976 [1], the public-key cryptography (PKC) is rapidly developed, which is served as the foundation of modern system and communication security. For example, the widely-used RSA, ElGamal and ECC have evolved as the most popular algorithms to implement public-key encryption

and digital signatures. The security of these cryptosystems depends on the hardness of certain problems in number theory such as the factoring problem (FP), the discrete logarithm problem (DLP) and elliptic curve problem [2]. However, these hard problems are proved to be very fragile when facing to the appearance of physical quantum computers that put Shor's quantum algorithm into reality [3]. Even though the quantum computers have not emerged on a large scale until now, it is urgent to research on new quantum-resistant cryptography algorithms to replace previous PKC, which are commonly named as Post-Quantum Cryptography (PQC).

PQC is usually divided into four classes: lattice-based, hash-based, code-based and multivariate-quadratic-equations-based cryptography [4]. In 2016, National Institute of Standards and Technology (NIST) has initiated a process to solicit, evaluate, and standardize the quantum-resistant public-key cryptography [5]. Among Round 1 submissions, 17 public-key encryption and key-establishment algorithms and 9 digital signature algorithms head to Round 2. The Supersingular Isogeny Key Encapsulation (SIKE) is the only one submission to NIST apart from other four classes, which is based on supersingular isogeny elliptic curves. And it is very promising and unique among all the Round 2 candidates.

For the supersingular isogeny-based cryptography, its debut is years later than those four classes aforementioned. It was first proposed by Jao and De Feo in 2011 [6], which relies on the conjectured hardness of finding isogenies between supersingular elliptic curves. Later, it was continuously optimized to improve its performance. For example, Faz-Hernández *et al.* proposed several algorithmic optimizations on both elliptic-curve and field arithmetic operations [7].

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Most importantly, they optimized the classic three-point ladder algorithm to compute elliptic-curve operation $P + [d]Q$, where P and Q are two points on the elliptic curve, and d is the secret scalar. Recently, it is increasingly prevailing to study how to implement and optimize the supersingular isogeny-based cryptography on embedded devices [8], [9], [10].

One of advantages of the isogeny-based cryptography is that the size of public and private key is very small so that it becomes possible to apply SIKE in many resource-constrained embedded devices. However, its computation speed is relatively slower than other four categories of post-quantum cryptosystems owing to the complex isogenies computations. In such computation, a rational algebraic map is found between two supersingular elliptic curves over a specific finite field. And thus the computations are moved from one elliptic curve to another. The scalar multiplication on elliptic curve, which is the core function in traditional elliptic curve cryptography (ECC), is also the kernel when computing the isogenies. To challenge this type of cryptography, the best attack ever known in mathematics utilizes quantum computers indeed, however, it still reaches to an exponential time complexity [11]. Therefore, it is also resistant to quantum computers in addition to above four kinds of PQC schemes.

The design of PQC mainly focuses on the resistance against both quantum computers and traditional mathematical analysis. However, it has been shown that side channel attack (SCA), which might be either possibly ignored by those designers of PQC, or realized to some extent yet not paid enough attention to its power of cryptanalysis.

Side-channel attack can extract the secret key of cryptographic devices by analyzing the side-channel leakage, such as power, electromagnetic emissions and timing etc. According to the types of leakage, it can be categorized into power analysis (PA) attack [12], electromagnetic attack (EMA) [13] and timing attack [14]. All of them are non-invasive attacks. The very first and most well-known SCA is the Differential Power Analysis (DPA) proposed in 1999 [12]. Besides DPA, Simple Power Analysis (SPA) [15] can also reveal the secret by only observing the power features. Compared to PA, EMA is more practical, convenient, and threatening because the electromagnetic emission is a three-dimensional signal, which can be located to the sensitive position of target devices in a contactless manner. Apart from SCA, some active faults can be injected to cryptographic devices to obtain the erroneous ciphertexts, which can also be used to deduce the secret key. This is called fault attack (FA) [16], most of which are semi-invasive.

Related Work. In terms of isogeny-based cryptosystem, there are several fault attacks in the past few years. For example, the first fault attack on this protocol was presented in [17], which aimed to change the base point to a random point via fault injections. Based on Ti's work, a loop-abort fault attack was proposed by G  lin, which exploited the iterative structure of secret isogeny computations [18]. However, side-channel attacks against this scheme are not that much in addition to three kinds of refined power analysis (RPA) attacks proposed by Koziel [19]. They are based on the zero-value representations in the context of quadratic extension fields and isogeny arithmetic. As for countermeasures, there are numerous methods to protect the scalar multiplication in ECC, such as [20], [21] and [22]. Yet, few countermeasures

are studied to protect the isogenies computation of SIKE from SCAs. Therefore, the security evaluation of SIKE under side-channel attacks is not enough up to now, and thus the research on the SCAs and countermeasures on the quantum-resistant SIKE algorithm will contribute a lot to the standardization of post quantum cryptography.

Contribution. Our contributions are listed as follows:

- We describe a thorough methodology to detect both horizontal and vertical side-channel leakage in the reference and optimized SIKE implementations, and a vertical vulnerability is identified in the optimized SIKE that DPA can theoretically target at. Especially, to the best of our knowledge, we are the first to propose the specific DPA method to attack the supersingular isogeny-based cryptography which is protected by Three-point Montgomery Differential Ladder algorithm.
- In order to prove that the vulnerability in question is indeed exploitable and put the theoretical DPA into practice, we migrate the optimized version of SIKE to a practical 32-bit ARM platform (Cortex A8), and set up an electromagnetic experimental platform. By analyzing the EM leakage, we conduct a practical differential electromagnetic attack (DEMA) on SIKE. And the experimental results show that the Multi-Bit DEMA can successfully reveal the private key with hundreds of samples.
- Through further experiments, we demonstrate that SIKE algorithms running on Gigahertz ARM processors can actually be attacked by using normal oscilloscopes at a Megahertz sampling rate for the first time. This reduces the attack difficulty for the adversary, and moves the SCA towards the practice with lower barriers.
- In order to strengthen the security of SIKE algorithm, we propose an efficient window-based countermeasure taking both security and performance into account. Compared to those countermeasures that are designed for ECC, our proposed method can resist the attack in this paper and probably other SCAs, such as refined power analysis. The window-based countermeasure has small time and memory costs, which carefully balances the trade-off between security and performance and will be more practical in related applications.

Organization. Section 2 gives the background of supersingular isogeny elliptic curves, SIDH and SIKE. Section 3 introduces the vulnerabilities detection on SIKE in terms of horizontal and vertical side-channel leakage. Section 4 shows a potential vertical vulnerability in the optimized SIKE and introduces DPA targeting on this vulnerability. Section 5 describes the practical DEMA based on the ARM platform, and shows the experimental results. Section 6 presents a window-based countermeasure to protect isogeny computation from SCA. And Section 7 concludes the paper.

2 BACKGROUND

In order to describe the supersingular isogeny Diffie-Hellman (SIDH) and supersingular isogeny key encapsulation

(SIKE) more clearly, some preliminary background will be introduced first, such as elliptic curve and isogeny. A more detailed description can be referred to [23], [24]. Then, our attack target—SIKE algorithm, which is based on SIDH protocol, will be briefly introduced.

2.1 Elliptic Curve Over Finite Fields

A basic finite field is defined as $\mathbb{F}_p$ with p elements, where p is a prime. The extension field of $\mathbb{F}_p$ is defined as $\mathbb{F}_q$ if q is a power of prime p , and the characteristic of $\mathbb{F}_q$ is defined as p . The finite fields applied in SIDH and SIKE are quadratic extension fields of a prime field $\mathbb{F}_p$, that is $\mathbb{F}_q$, where $q = p^2$ and $p = 2^{e_2}3^{e_3} - 1$ with fixed public parameters e_2 and e_3 . This extension field is formed as $\mathbb{F}_{p^2} = \mathbb{F}_p(i)$ where i is a representation in complex field with $i^2 + 1 = 0$ [25].

Elliptic Curve. An elliptic curve E over $\mathbb{F}_q$ is a set of points that satisfy Eq.(1):

$$E(\mathbb{F}_q) = \{(x, y) \in \mathbb{F}_q : y^2 = x^3 + ax + b\} \cup \{\mathcal{O}\}, \quad (1)$$

where a and b are two coefficients over $\mathbb{F}_q$, and $\mathcal{O}$ is the point at infinite (∞). This kind of elliptic curve with standard form is commonly called as Weierstrass elliptic curve [26].

Montgomery Elliptic Curve. A Montgomery curve, used in SIDH and SIKE, is a class of special elliptic curve. It is denoted as a set of points which satisfy Eq. (2), where $a, b \in \mathbb{F}_q$ and $b(a^2 - 4) \neq 0$.

$$E_{a,b}(\mathbb{F}_q) = \{(x, y) \in \mathbb{F}_q : by^2 = x^3 + ax^2 + x\} \cup \{\mathcal{O}\}. \quad (2)$$

The Montgomery elliptic curve was first proposed to speed up the computations on elliptic curve [27], so it is more often used in ECC practice. Besides, the y -coordinate of the point $P = (x_P, y_P)$ in Montgomery elliptic curve can be ignored because it can be recovered by other computations when required [28]. Thus only the x -coordinate x_P is utilized in the computations on Montgomery elliptic curve.

Supersingular Elliptic Curve. Specially, an elliptic curve $E(\mathbb{F}_q)$ with $q = p^k$ is called as supersingular curve if $\#E(\mathbb{F}_q) \equiv 1 \pmod{p}$, where p is a prime and $k \in \mathbb{Z}$. $\#E(\mathbb{F}_q)$ is defined as the order of an elliptic curve E over a finite field $\mathbb{F}_q$, which is the number of points in E including the infinite point $\mathcal{O}$. The application of supersingular curves is deemed to reduce the computational complexity of elliptic curve discrete logarithm problem [29]. When facing the challenge from quantum computers, it is showed that those cryptographic algorithms based on ordinary curves are insecure. However, new designs based on supersingular ones might have much more security strength [23], and thus the supersingular isogeny-based cryptography is able to head to the second round of NIST's PQC competition.

Scalar Multiplication in Projective Coordinate. Two main point operations over elliptic curve are point doubling $([2]P = (x_{[2]P}, y_{[2]P}))$ and point addition $(P + Q = (x_{P+Q}, y_{P+Q}))$. A scalar multiplication $[d]P = (x_{[d]P}, y_{[d]P})$, which is the core operation of ECC and SIKE, can be computed via series of point doublings and additions. d is the so-called scalar which is a key-dependent target for most attacks. Moreover, projective coordinate is suggested to avoid the inversion operations and improve the computation performance [30].

The projective form of the Montgomery curve is defined as $E_{A:B:C}$ and shown in Eq. (3), where $a = A/C$ and $b = B/C$.

$$E_{A:B:C} : By^2 = Cx^3 + Ax^2 + Cx. \quad (3)$$

Therefore, the affine coordinate (x, y) can be transformed into projective coordinate $(X : Y : Z)$, where $x = X/Z$ and $y = Y/Z$ with $Z \neq 0$. The infinite point $\mathcal{O}$ can be written as $(X : Y : Z) = (0 : 1 : 0)$. As the Y -coordinate can be recovered using Okeya-Sakurai strategy [28], it can be omitted for the purpose of convenience. Thus this point can be represented as $(X : Z)$. Correspondingly, the doubling and addition operations are denoted as $(X_{[2]P} : Z_{[2]P})$ and $(X_{P+Q} : Z_{P+Q})$, respectively.

2.2 Isogeny

Isogeny theory describes the relationship of various elliptic curves. Let E_1 and E_2 be two elliptic curves over $\mathbb{F}_q$. Then, an isogeny $\phi: E_1 \rightarrow E_2$ is a non-constant rational mapping over $\mathbb{F}_q$, and ϕ is also a group homomorphism from $E_1(\mathbb{F}_q)$ to $E_2(\mathbb{F}_q)$. E_1 and E_2 are isogenous over $\mathbb{F}_q$ if and only if $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$. It can be concluded that isogenies are group homomorphisms, so they are “morphisms” in the sense of both algebraic geometry and group theory [23].

The kernel of an isogeny is defined as the set of points on the curve that can satisfy Eq.(4)

$$\ker(\phi) = \{P \in E_1 : \phi(P) = \mathcal{O}\}. \quad (4)$$

The isogeny computation can be obtained from its kernel, and it is common to use the kernel of an isogeny to construct the cryptographic core in SIDH [31]. For any subgroup $H \subseteq E_1$, there is a unique elliptic curve E_2 with an isogeny: $\phi: E_1 \rightarrow E_2$ such that $\ker(\phi) = H$. Both SIDH and SIKE use isogenies among supersingular elliptic curves instead of ordinary ones for security reasons, and these isogenies over subgroups are always 4-isogenies and 3-isogenies.

2.3 Supersingular Isogeny Diffie-Hellman Protocol

Classic Diffie-Hellman protocol agrees on a shared key on an insecure communication channel. Similarly, SIDH constructs the shared key by computing the j -invariant of two isomorphic supersingular elliptic curves E_A and E_B generated by two entities Alice and Bob. j -invariant is calculated as $j(E_{a,b}) = \frac{256(a^2-3)^3}{a^2-4}$.

A starting supersingular elliptic curve E_0 is defined over a finite field $\mathbb{F}_{p^2}$, where p is a large prime with the form of $p = (l_A)^{e_A}(l_B)^{e_B}f \pm 1$. l_A and l_B are small primes. e_A and e_B are positive integers. And f is a small cofactor. Note that in SIKE, $l_A = 2, l_B = 3, f = 1$, respectively. The order of E_0 is determined: $\#E_0 = (l_A^{e_A}l_B^{e_B}f)^2 = (2^{e_A}3^{e_B})^2$. Then two pairs of points $(P_A, Q_A), (P_B, Q_B)$ are selected to generate two subgroups $E[l_A^{e_A}]$ and $E[l_B^{e_B}]$. It should be noted that the prime p , the initial curve E_0 and those generated points $(P_A, Q_A), (P_B, Q_B)$ are all public parameters.

Alice chooses two random elements $m_A, n_A \in \mathbb{Z}_{l_A^{e_A}}$ privately, and computes the isogeny: $\phi_A: E_0 \rightarrow E_A$ with kernel $\ker(\phi_A) = \langle [m_A]P_A + [n_A]Q_A \rangle = \langle R_A \rangle$. Then she calculates $\phi_A(P_B)$ and $\phi_A(Q_B)$, and sends them to Bob together with her curve E_A . When Bob receives $(E_A, \phi_A(P_B), \phi_A(Q_B))$

from Alice, he computes $\phi_{BA} : E_A \rightarrow E_{BA}$ with $\ker(\phi_{BA}) = \langle [m_B]\phi_A(P_B) + [n_B]\phi_A(Q_B) \rangle = \langle \phi_A(R_B) \rangle$. Similarly, Alice obtains the isogeny $\phi_{AB} : E_B \rightarrow E_{AB}$, whose kernel is $\langle [m_A]\phi_B(P_A) + [n_A]\phi_B(Q_A) \rangle = \langle \phi_B(R_A) \rangle$. Finally, the agreed key can be obtained based on ϕ_{BA} and ϕ_{AB} .

In order to compute the secret kernel function more efficiently, $R = [m]P + [n]Q$ is usually simplified to $R = P + [d]Q$ with $d = m^{-1}n$, assuming that m or n is invertible modulo the order of the group.

2.4 Supersingular Isogeny Key Encapsulation Protocol

Supersingular isogeny key encapsulation (SIKE) is built on the top of SIDH key-exchange construction, and implements public-key encryption and key encapsulation mechanism [32].

When Bob initiates a communication with Alice, he generates a random key pair (sk_B, pk_B) . He sends public key pk_B to Alice and keeps private key sk_B by himself. After Alice receives pk_B , she produces a random entry m , and encrypts it with pk_B to obtain a ciphertext ct and a shared key $\mathcal{K}$. Then ct is sent back to Bob. When the original initiator Bob receives ct , he decrypts it with his own private key sk_B . Therefore, Bob can recover the random entry m used by Alice and generate the shared key $\mathcal{K}$. Subsequently, a secure transmission is built between two parties using this symmetric key. Details can be referred to the specification of SIKE [32].

3 VULNERABILITIES DETECTING ON QUANTUM-RESISTANT SIKE ALGORITHM

In this section, we will analyze the implementation security of SIKE submitted to NIST using a thorough methodology to detect vulnerabilities, both in terms of horizontal and vertical side-channel leakage. Horizontal leakage means that the leakage exists in x -axis direction, which is often related to the running time [33]. The horizontal leakage usually can be observed via a single trace, and it can be exploited by simple power analysis and timing attacks. In contrast, vertical leakage must involve different inputs and the difference only exists in y -axis direction [33]. In general, it cannot be observed directly, instead some statistical methods have to be applied, such as differential power analysis and correlation power analysis [34]. It can be demonstrated in this paper that even if the SIKE algorithm is quantum-resistant, it can still be threatened by side-channel attacks if it has horizontal or vertical vulnerabilities.

3.1 SIKE Implementations in NIST Submission

The SIKE algorithm submitted to NIST mainly includes three types of implementations: reference, optimized, and additional ones. The reference implementation exploits the Montgomery Double-and-Add Chain (MDAC) algorithm in portable C language. And it is already mentioned in [32] that such vanilla implementation is vulnerable to SPA. The optimized implementation utilizes the Three-point Montgomery Differential Ladder (TMDL) algorithm to defend against SPA. The additional version contains several software implementations based on 64-bit ARM and AMD64 platforms using assembly language, and hardware implementation in VHDL. We mainly focus on the vulnerabilities

in reference and optimized versions, because additional ones are implemented all through TMDL and actually just extensions of optimized implementation.

In both reference and optimized implementations, there are several different parameters. In the first submission to NIST, there are three parameters: SIKE_{p503} , SIKE_{p751} and SIKE_{p964} according to the bit length of the prime field characteristic p . In order to meet the standard defined by NIST, two other parameters are added: SIKE_{p434} and SIKE_{p610} , while SIKE_{p964} is removed in Round 2. Through careful analysis, these four different parameters meet the respective security requirements of NIST's categories 1, 2, 3 and 5 [32]. And only their public/private key lengths are different.

Since the parameter will not affect the result of our vulnerability detection, SIKE_{p503} is selected as an example in this paper, which is included in both Round 1 and 2. However, all these methodologies can be extended to SIKE_{p434} , SIKE_{p610} and SIKE_{p751} quite straightforwardly.

3.2 Horizontal Leakage Detection

In order to detect the horizontal leakage in both reference and optimized implementations, a powerful source code analyzer is exploited, which can perform static analysis with tainting propagation of sensitive variables across the source codes and detect leakage patterns [35]. Practically, this tool has been applied to evaluate the crypto libraries, such as MbedTLS [36]. Originally it is built to identify the vulnerabilities of cache-timing attacks. However, this tool can also be used to detect the leakage in horizontal direction, which can be exploited by cache-timing attacks, simple power analysis and other attacks. The horizontal leakage points we mainly focus on include the secret-dependent conditional instructions and jump instructions, which may cause the non-constant running time.

In [35], the authors have analyzed all the candidates submitted to NIST in Round 1 except the SIKE algorithm. Therefore in this paper, a complete horizontal leakage detection is performed for SIKE using that source code analyzer. The sensitive variable in SIKE algorithm is determined as the key-related scalar d in scalar multiplication. Then whether some conditional instructions will depend on this sensitive variable or not is determined. After vulnerabilities detection on SIKE, several horizontal leakages are found in the reference version but not in the optimized one. In other words, the reference implementation runs in non-constant time, which is consistent with the claims from algorithm designer [32]. And the optimized implementation runs in constant time due to the application of Three-point Montgomery Differential Ladder algorithm. Thus the measured power or electromagnetic traces are well aligned, which makes the optimized SIKE resistant to cache timing attack and simple power analysis.

3.3 Vertical Leakage Detection

Since the optimized version of SIKE runs in constant time and has no horizontal vulnerabilities, it is naturally possible to try the detection on the vertical sources of leakage.

One classic method to detect such vertical leakage is based on Welch's t -test, which was first adopted by Goodwill to

TABLE 1
Three-Point Differential Ladder With $d' = (10101)_{bin} = (21)_{dec}$

Projective coordinates	Init	$d_0(\text{LSB})$	d_1	d_2	d_3	d_4
R_0	Q	$\mathbf{1}$	0	1	0	1
R_1	P	$[2]Q$	$[4]Q$	$[8]Q$	$[16]Q$	$[32]Q$
R_2	$Q - P$	$P + Q$	$P + Q$	$P + [5]Q$	$P + [5]Q$	$P + [21]Q$
		$Q - P$	$[3]Q - P$	$[3]Q - P$	$[11]Q - P$	$[11]Q - P$

evaluate SCA in 2011 [37]. Through varying all the bits of AES key at only one position and tracking differences in the traces, the side-channel vulnerabilities of a cryptographic device can be detected. Later this method was extended to public cryptosystems such as RSA [38]. Furthermore, t -test was improved by Becker and renamed as Test Vector Leakage Assessment (TVLA) [39].

However, there are several worries if applying t -test (or TVLA) to detect vertical vulnerabilities for SIKE at early states. First, t -test is a method for post-processing after all traces are collected from devices. However it is not good for pre-processing when such evaluation is conducted at the source code level. Note that we would like to detect the possible leakage points first before we fulfill the physical experiments. Second, the length of scalar and the whole duration of SIKE are very long, which will cause t -test extremely slow. Therefore, these methods are not used in our work. Instead, another type of theoretical vertical vulnerability analysis based on algorithm and source code is conducted in Section 4.

4 VERTICAL LEAKAGE AND POWER ANALYSIS

Section 3 shows that there are horizontal vulnerabilities in the reference implementation of SIKE, which is already known to us. So it is not considered in this paper. In addition, Section 3 proves that the optimized SIKE implementation runs in constant time and resistant to SPA. However, this section will present a potential vertical vulnerability of isogenies computation of SIKE, which is based on Three-point Montgomery Differential Ladder algorithm. In the meanwhile, a theoretical Differential Power Analysis (DPA) targeting on this vulnerability will be elaborated.

4.1 Three-Point Montgomery Differential Ladder (TMDL)

The optimized implementation uses efficient algorithms for isogenies computations based on projective coordinates on supersingular Montgomery curve. Its core function is still the scalar multiplication $P + [d]Q$, which is implemented with the recent and improved *Three-point Montgomery Differential Ladder* (TMDL) [7].

Algorithm 1 shows the TMDL algorithm, which is realized in projective coordinates. Compared to traditional double-and-add algorithm, it not only improves the resistance against SPA, but also speeds up the computations significantly, because it calculates $P + [d]Q$ instead of $[d]Q$ merely. In addition, a Montgomery point in affine coordinate (x, y) is transformed to the projective tuple representation $(X : Z)$ without Y coordinate to avoid inversions, where $x = X/Z$.

In Algorithm 1, when the first bit d_0 is 1, $(X_0 : Z_0)$ is doubled and $(X_0 : Z_0) + (X_1 : Z_1)$ is computed, which means that $[2]Q$ and $P + Q$ are obtained. Otherwise, $(X_0 : Z_0)$ is

doubled yet $(X_0 : Z_0) + (X_2 : Z_2)$ is computed. That is, $[2]Q$ and $Q + (Q - P)$ are derived. Finally, $(X_1 : Z_1)$ returns $P + [d]Q$ when all the bits of d are traversed. Note that TMDL scans all the key bits from right (LSB) to left (MSB). The scalar will be padded with zeros on the left if its length is less than l .

Algorithm 1. Three-point Montgomery Differential Ladder in Projective Coordinates to Compute $P + [d]Q$

Input: $d = (d_{l-1}, \dots, d_0)$, (x_P, x_Q, x_{Q-P}) , $(a_{24}^+ : 1)$

Output: $(X_{P+[d]Q} : Z_{P+[d]Q})$

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1:  $R_0 = (X_0 : Z_0) \leftarrow (x_Q : 1)$ ,  $R_1 = (X_1 : Z_1) \leftarrow (x_P : 1)$ ,
    $R_2 = (X_2 : Z_2) \leftarrow (x_{Q-P} : 1)$ ;
2: for  $i = 0; i < l; i + \mathbf{do}$ 
3:   if  $d_i == 1$  then
4:      $((X_0 : Z_0), (X_1 : Z_1)) \leftarrow \text{xDBLADD}((X_0 : Z_0), (X_1 : Z_1), (X_2 : Z_2), (a_{24}^+ : 1))$ ;
5:   end
6:   else
7:      $((X_0 : Z_0), (X_2 : Z_2)) \leftarrow \text{xDBLADD}((X_0 : Z_0), (X_2 : Z_2), (X_1 : Z_1), (a_{24}^+ : 1))$ ;
8:   end
9: end
10: return  $(X_1 : Z_1)$ 

```

In this algorithm, the xDBLADD function achieves the point doubling and differential point addition simultaneously, where a_{24}^+ is a constant used for differential computation. This function will be always executed regardless of whether the bit d_i is 1 or 0. Moreover, the inner implementations of xDBLADD function are with 19 fixed instructions without any branches and conditional jumps [32]. Therefore, this Three-point Montgomery Differential Ladder runs in constant time and it is designed as an effective countermeasure against SPA, which meets the descriptions of Section 3.

4.2 Vertical Vulnerability in TMDL and Differential Power Analysis

In order to find the vertical leakage to be targeted by DPA, two different secrets d are taken into consideration: one is $d' = (10101)_{bin} = (21)_{dec}$ whose least significant bit (LSB) is 1; the other is $d'' = (11010)_{bin} = (26)_{dec}$ whose LSB is 0. Tables 1 and 2 present the computation processes of three-point differential ladder algorithm for d' and d'' , respectively. R_0, R_1, R_2 are three temporary variables in projective coordinate.

In Table 1, to compute $P + [d']Q$, point Q is always doubled in each iteration thus $[2]Q, [4]Q$ and $[8]Q \dots$ are obtained. Then $P + Q$ is calculated since $d'_0 = 1$ and $[2]Q + (Q - P)$ is computed since $d'_1 = 0$. In the same way, the final result $P + [21]Q$ is obtained, which is colored as blue in Table 1.

TABLE 2
Three-Point Differential Ladder With $d'' = (11010)_{bin} = (26)_{dec}$

Projective coordinates	Init	$d_0(\text{LSB})$	d_1	d_2	d_3	d_4
R_0	Q	0	1	0	1	1
R_1	P	$[2]Q$	$[4]Q$	$[8]Q$	$[16]Q$	$[32]Q$
R_2	$Q - P$	P	$P + [2]Q$	$P + [2]Q$	$P + [10]Q$	$P + [26]Q$
		$[2]Q - P$	$[2]Q - P$	$[6]Q - P$	$[6]Q - P$	$[22]Q - P$

In Table 2, intending to compute $P + [d'']Q$, point Q is still doubled in each round as in Table 1. Since $d''_0 = 0$, $Q + (Q - P)$ is first calculated instead of $Q + P$. And due to $d''_1 = 1$, $[2]Q + P$ is computed subsequently. The computation is recursively executed, and the result $P + [26]Q$ is obtained eventually, which is also marked in blue in Table 2.

In order to identify the vertical leakage, we take a close look at both tables, and the column of d_0 (LSB) draws our attention where $d'_0 = 1$ in Table 1 and $d''_0 = 0$ in Table 2. When $d_0 = d'_0 = 1$, three temporary values are $\{[2]Q, P + Q, Q - P\}$ for R_0, R_1, R_2 , respectively. When $d_0 = d''_0 = 0$, they are $\{[2]Q, P, [2]Q - P\}$ for R_0, R_1, R_2 as shown in Table 2. *The most important observation is that: whether $P + Q$ or $2[Q] - P$ is calculated is dependent on whether d_0 is 1 or 0.* According to SCA theory, this difference in d_0 is associated with the power dissipation and can be recognized by some statistical analysis if the number of samples is enough. Therefore, the Three-point Montgomery Differential Ladder algorithm shows vertical leakages and it is vulnerable to DPA attack. These calculations of $P + Q$ and $[2]Q - P$ can serve as the selection function for DPA practice. It should be noted that the Montgomery points $P + Q$ and $[2]Q - P$ are computed in the form of projective coordinates: $(X_{P+Q} : Z_{P+Q})$ and $(X_{[2]Q-P} : Z_{[2]Q-P})$. Therefore in this paper, the projective x -coordinates, X_{P+Q} and $X_{[2]Q-P}$, are chosen as our intermediate values for selection functions in DPA.

Because the instantaneous power consumption is closely related to the processed data, the power features will be theoretically correlated to any specific bit of point $P + Q$ or $[2]Q - P$ when it is processed. In other words, when point $P + Q$ is computed, the power consumption will show certain difference according to any bit of $P + Q$ to some extent. In the opposite, no divergence will be observed if point $P + Q$ is not processed. In summary, there exists vertical differences when the secret scalar d varies. Thus it is possible to extract the bits of private scalar by guessing which point is computed.

If the first bit $d_0 = 1$, $P + Q$ is processed in SIKE, and power consumption is thus correlated with any specific bit of $P + Q$. Otherwise if $d_0 = 0$, power consumption will be correlated with the bits of $[2]Q - P$ instead of $P + Q$. Assuming that the SIKE algorithm is repeated N times with random $P_1, P_2, \dots, P_N$ and $Q_1, Q_2, \dots, Q_N$ as inputs, and the private and public key are fixed every time. In addition, define $C_i(t)$ (t is the moment in time) as the power consumption of each execution of SIKE with $1 \leq i \leq N$. Note that intermediate values X_{P+Q} and $X_{[2]Q-P}$ contain 126 bytes (1008 bits), so that two kinds of analysis methods can be considered: Single-Bit DPA and Multi-Bit DPA.

As for Single-Bit DPA, all the power traces are divided into two sets according to whether the specific bit s_j ($0 \leq j \leq 1007$) of intermediate values is 1 or 0. s_j is one bit of either X_{P+Q} or $X_{[2]Q-P}$. Then the differential power

function $\delta(t)$ between the power traces $C_i(t)$ and this specific bit s_j can be computed as in Eq. (5). $\text{num}(s_j = 1)$ denotes the total number of traces whose specific bit $s_j = 1$ for intermediate values X_{P+Q} or $X_{[2]Q-P}$.

$$\delta(t) = \frac{\sum_{i=1}^N (C_i(t) | s_j = 1)}{\text{num}(s_j = 1)} - \frac{\sum_{i=1}^N (C_i(t) | s_j = 0)}{\text{num}(s_j = 0)}. \quad (5)$$

A power trace $C_i(t)$ can be partitioned into Set 0 if the certain bit s_j of intermediate value X_{P+Q_i} is 0 ($1 \leq i \leq N$), while it is classified into Set 1 if s_j is 1. Then if the average power consumption of Set 0 shows a significant difference from Set 1, $P_i + Q_i$ is processed in SIKE decapsulation and the first bit d_0 of private key can be deduced as 1. Otherwise, d_0 is inferred as 0 if there is no difference between Set 0 and 1. Similarly, the average power of Set 1 for those points $X_{[2]Q_i-P_i}$ with $s_j = 1$ will be different from Set 0 for which $s_j = 0$ if $d_0 = 0$, while no divergence will exist if $d_0 = 1$.

As for Multi-Bit DPA, all the 1008 bits are considered, instead of only one bit. Therefore, its performance is better than Single-Bit DPA due to comprehensive contribution from each individual bit of intermediate values. The new function $g(t)$ is the sum of $\delta(t)$ in Single-Bit DPA for all the 1008 bits ($s_0, \dots, s_{1007}$), which is defined in Eq. (6). Then $g(t)$ of intermediate value X_{P+Q} or $X_{[2]Q-P}$ will present a more distinguishable peak according to $d_0 = 1$ or 0 during their execution, respectively.

$$g(t) = \sum_{j=0}^{1007} \delta(t) | s_j. \quad (6)$$

After the first bit d_0 is recovered, the second bit d_1 can also be revealed in the same way just with different intermediate values, which can be determined as in Tables 1 and 2. For example, supposing the first bit $d_0 = 0$, then $[2]Q + P$ will be computed when $d_1 = 1$, while $[4]Q - P$ will be calculated when $d_1 = 0$. So $[2]Q + P$ and $[4]Q - P$ should be selected as the intermediate values when recovering d_1 . Finally, all of the rest scalar bits can be recovered one by one, recursively.

5 PRACTICAL ELECTROMAGNETIC ATTACK ON SIKE BASED ON ARM PLATFORM

After identifying the vertical vulnerability of Three-point Montgomery Differential Ladder algorithm of SIKE and elaborating the DPA attack at the theoretical level, in this section, we will prove that the vulnerability in question is indeed exploitable for a successful key recovery in physical environments. A practical electromagnetic attack (EMA) based on ARM platform is conducted in our experiments and will be described in the following.

5.1 Moving From Power to Electromagnetic Attack

Even though power analysis and electromagnetic analysis are similar at the theoretical level, the practices of these two attacks are quite different, if taking the leakage measurement into consideration. In terms of power analysis attack, the power consumption needs to be collected via a small resistor between the power supply and the ground. This setup is very common, however, it is mainly applied to two scenarios. One is the customized breadboard where a measurement circuit has to be assembled including the required resistor. The other is the SASEBO/SAKURA set where the standard evaluation board is already configured with the resistor and its jumper. In practical situations, the cryptographic embedded devices may not provide such a convenient or flexible setup, so that the adversary Eve can not measure the power leakage without invading or breaking the devices. Moreover, the collected power signals may be influenced by other noise, because the power is supplied for all the components on the device, not just for the target processor. By contrast, the electromagnetic radiation is a signal of three dimensions in space, and the functional unit running those sensitive operations on the target device can be positioned by the EM probe accurately in a contactless manner. So the EM measurements can precisely reflect the actions in a general processor or microprocessor, and the influence of noise can be reduced as much as possible. Therefore, EM attack is a more practical and convenient method, which may cause great threats to the physical implementations of PQC.

5.2 Experimental Setup

Hardware Platform. The FriendlyARM series development board—Smart210 is selected as the host device, which carries with Samsung S5PV210 as the core processor. S5PV210 is a 32-bit RISC low power and high performance microprocessor for mobile phones and general applications. It integrates the ARM Cortex-A8 core, which implements ARM V7-A Instruction sets. In addition, the maximum frequency of S5PV210 is up to 1 GHz, which satisfies the speed requirement of slow PQC algorithms. Furthermore, the peripherals include 512M DDR2 RAM and 2G MLC NAND Flash, which meet the storage requirement of large public and private key size of all the PQC algorithms including SIKE.

The electromagnetic measurement station of Riscure is used to fix and move the EM probe. The corresponding EM probe used for leakage collection is also from Riscure. For the sake of accuracy, the high sensitivity Probe—EM Probe 340 HS is selected in our experiments. The mobile station can move the EM Probe automatically in X-Y-Z three dimensions to detect the hotspot of strongest electromagnetic radiation, and then the probe can pick up the EM-fields signals and convert them into AC signals. Fig. 1 shows the EM measurement platform used in our experiments.

Furthermore, a digital oscilloscope DSAV164A from Keysight, with 16G maximum bandwidth and 80GSa/s (samples per second) maximum sampling rate, is used to sample signals received from the EM Probe. In our experiments, due to the maximum operating frequency of target device up to 1 GHz, a sampling rate at 2.5GSa/s is determined first so as to meet the requirement of Nyquist sampling theorem.

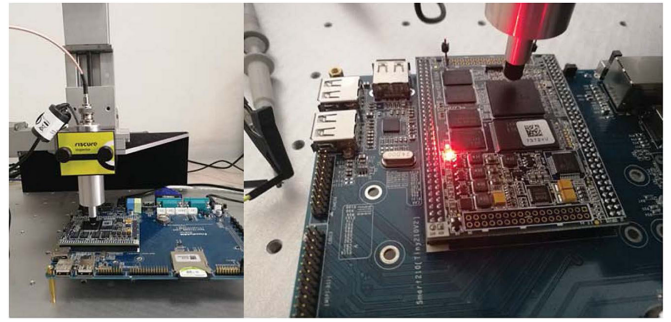


Fig. 1. ARM-based EM measurement platform.

Software Support. A Linux OS with YAFFS2 file system is running on the ARM-based platform. The kernel version is Linux-3.0.8. We use this OS to port the source code directly and ensure that there is no modification to the code. On other platforms, such as ARM Cortex-M, without such Linux OS on the board, the code might need some modification to implement. In that case, it is difficult to judge whether further vulnerabilities come from the newly introduced modification or from the original implementation instead. In that case, it is difficult to judge whether further vulnerabilities come from the newly introduced modification or from the original implementation instead. A cross compilation chain arm-linux-gcc-4.5.1-v6-vfp is installed in a desktop PC with Ubuntu 18.04. The source code of SIKE is cross-compiled and migrated to ARM platform. In addition, a serial port communication program is written in Python to control the execution of SIKE and manage oscilloscope storage. After EM traces are collected, corresponding cryptanalysis is conducted in MATLAB 2018a.

5.3 DEMA on Optimized Implementation of SIKE

The optimized implementation of SIKE with Three-Point Montgomery Differential Ladder runs in constant time and has no horizontal leakage, however, it still presents a potential vertical vulnerability and maybe threatened by DPA as analyzed in previous section. Therefore, Differential Electromagnetic Attack (DEMA) is put into practice to verify the credibility of theoretical analysis.

When the optimized implementation of SIKE is executed in ARM development board, 5000 EM traces at a sampling rate of 2.5 Gsa/s are collected with random messages input. For the purpose of illustration, only the interested part around Three-Point Montgomery Differential Ladder instead of the whole SIKE algorithm is measured. The bad side is that the size of traces under such a sampling rate is very large and not easy to process. However, the good side is that the analysis can be conducted in a fashion of bit by bit, which requires a small truncation from the entire trace each time. All the EM traces are preprocessed with the static alignment method [15] to improve the attack accuracy.

In order to evaluate the results of DEMA, two metrics are proposed: the minimum traces to disclose the key bit (MTD) and the maximum value of peak (MVP). When the number of traces is increased up to a certain value, the correlation of correct guess will be always greater than that for the wrong guess. This value is defined as MTD. It is desired to be as small as possible for a successful attack. In addition, there will exist

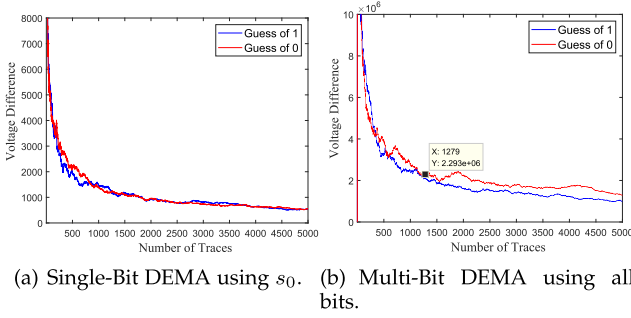


Fig. 2. Attacking results of d_0 using single-bit and multi-bit DEMA on the optimized implementation of SIKE.

a peak at the moment of executing the key-dependent operations if the attack is successful. The maximum amplitude of this peak is denoted as MVP. The attack is easier to conduct if the MVP is much larger.

5.3.1 Single-Bit DEMA

As described in Section 4, any single bit of intermediate values can be exploited in Single-Bit DPA to carry out the attack theoretically. However, in Fig. 2a, Single-Bit DEMA based on s_0 , the first bit of intermediate values, does not work well as expected. It can not deduce whether d_0 is 1 or 0 even if the number of traces is raised to 5000. This also happens to other bits such as s_1 , s_2 and so on. The attack power from each single bit s_j ($0 \leq j \leq 1007$) is significantly mitigated by the existence of noise from different aspects, such as environment and measurement noise, etc.

5.3.2 Multi-Bit DEMA

The contribution of the attack power from each single bit, however, can still be accumulated and used to extract the scalar successfully. Fig. 2b shows how voltage difference corresponds to the number of traces in Multi-Bit DEMA. In Fig. 2b, the marked point (1279, 2.293×10^6) indicates that MTD is 1279, which means that about 1300 EM traces are enough to extract $d_0 = 0$ in this practical attack.

Since the first bit d_0 is confirmed as 0, in order to confirm the robustness of the attack, the second bit d_1 is also analyzed where $d_1 = 0$ indeed. Looking back Table 2 in Section 4, when $d_1 = 1$, $R_0 + R_1 = [2]Q + P$ will be computed. Otherwise, $R_0 + R_2 = [2]Q + ([2]Q - P) = [4]Q - P$ is processed. Therefore, $[2]Q + P$ and $[4]Q - P$ can be used as the intermediate values to attack d_1 . Fig. 3 shows the results of DEMA when the first two bits of private key are analyzed. The EM traces are truncated for each bit separately and then illustrated together for the convenience of comparison.

Fig. 3a demonstrates the correlation function $g(t)$ of DEMA when d_0 is analyzed. Intermediate values $P + Q$ and $[2]Q - P$ are used to classify the electromagnetic traces. For Guess 1 marked in blue, there is no significant pattern that can be detected. Most of the voltage difference values are within the range $[-0.75 \times 10^6, 0.75 \times 10^6]$, indicating that the guess is incorrect. For Guess 0 marked in red, there is a significant voltage difference for the first 1.5×10^6 points, ranging in $[-1.1 \times 10^6, 1.1 \times 10^6]$, showing that Guess 0 is correct for d_0 .

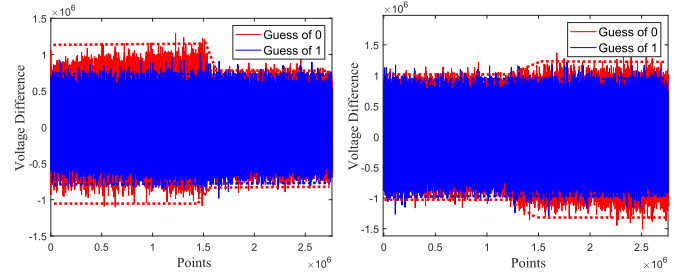


Fig. 3. DEMA results of attacking the first two bits of d .

Similarly, Fig. 3b shows the attack result for d_1 through $[2]Q + P$ and $[4]Q - P$. For Guess 0 marked in red, the differentiable part appears at the point 1.5×10^6 afterwards. Its voltage difference values are within $[-1.3 \times 10^6, 1.3 \times 10^6]$, which is still detectable and implies that Guess 0 is correct for d_1 .

Note that both Fig. 3a and 3b are analyzing the same batch of traces but with different intermediate values. The corresponding results show the different locations for the correct guesses ($d_0 = 0$ and $d_1 = 0$). This is also consistent with the fact that multiplication operation is actually scanning the scalar d bit by bit. And the analysis can also be conducted bit by bit.

Furthermore, the peaks in Fig. 3a and 3b are not as that significant as those in traditional attacks. The reason behind can be explained that the actual computations of intermediate values in SIKE are fulfilled by those low-level instructions. The number of those underlying instructions is quite a lot, each of which holds the power-data correlations. Besides, the operations might be executed on different physical components that are in the different positions of the processor. Therefore, the maximum correlation is still detectable, however, it is widely spread on all those related operations thus not that differentiable.

5.4 Downsampling DEMA on SIKE

So far, the attack on SIKE is successfully conducted at a sampling rate of 2.5GSa/s supported by a very powerful digital oscilloscope. However, it should be emphasized that 5000 electromagnetic samples eat nearly 30GB storage space at this sampling rate, and the speed of analysis is very slow owing to the large number of points in each sample. So, it is thought to be very expensive to launch such an attack in practice, when considering the scope, the space, and the analysis speed.

However, as shown in Fig. 3, each operation is represented with millions of points, implying that the execution of multiplication is extremely slow, even for scanning one bit. This enlightens our doubt whether we really need such a high sampling rate. Since DEMA is verified, it is interesting to know to what extent a lower sampling rate can still manage to work. Therefore, the electromagnetic traces are measured via the same oscilloscope but at a sampling rate from 2.5GSa/s down to 100MSa/s, with a down step of 100MSa/s. Then Multi-Bit DEMA is conducted for each step and the attack results including MTD and MVP are recorded.

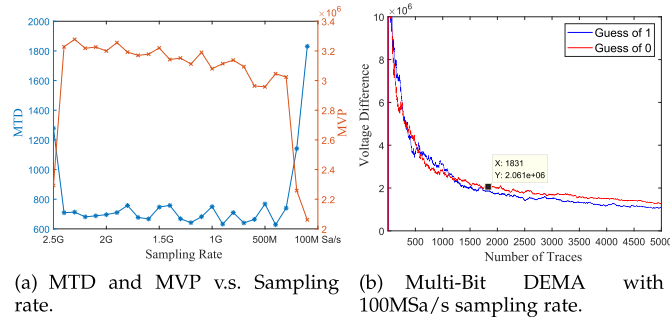


Fig. 4. DEMA results when considering different sampling rates from 2.5GSa/s down to 100MSa/s.

Fig. 4a demonstrates the relationship between the attack performance and different sampling rates. The blue and red curves depict how MTD and MVP change with the sampling rate, respectively. Surprisingly, we can see that the performance is actually improved when the sampling rate is reduced from 2.5GSa/s. Especially, only 628 samples are required to conduct a successful attack when the sampling rate is 400MSa/s. In addition, the performance starts to get worse when the sampling rate decreases to below 300MSa/s. As shown in Fig. 4b, 100MSa/s is still a sampling rate that can manage to attack, whose MTD has increased to 1831 and MVP decreases to 2.061×10^6 . However, in our experiments, DEMA can not succeed when the sampling rate is under 100MSa/s.

In previous works, the choice of the sampling rate is expected to satisfy the requirement of Nyquist Theorem When conducting SCA on high speed processors. For example, the AES algorithm was attacked using DPA on ARM Cortex-A8 processor running at 1 GHz. However, the sampling rate is up to 6 GSa/s [40]. But in this paper, the required sampling rate can be down to Megahertz, even if SIKE runs on a Gigahertz processor. This discovery may inspire us to attack other PQC algorithms at a smaller sampling rate even if the algorithms run on a processor with very high speed. Therefore, the attacks on complex PQC algorithms, such as SIKE, can be more easily put into practice, taking the experimental equipments into account.

6 COUNTERMEASURES

Throughout this paper, a theoretical vertical vulnerability in the optimized SIKE protocol is detected and it is demonstrated to be vulnerable under power analysis or electromagnetic attacks, especially for the scalar multiplication in isogenies computations. There are some countermeasures to protect scalar multiplication as for elliptic curve cryptography (ECC). For example, Coron suggested three types of countermeasures to prevent DPA [20]. Similarly, Joye and Ciet also proposed several randomization methods [21], [22].

However, these countermeasures used for ECC may be not migrated to SIKE algorithm directly, because it is based on a supersingular elliptic curve over extension field $\mathbb{F}_{p^2}$ instead of ordinary curve over prime field $\mathbb{F}_p$. Note that a countermeasure designed to prevent DPA may lead to other vulnerabilities. Therefore, it is worth proposing our new solution targeting at this vertical vulnerability and evaluating it against existing implementation attacks. In addition,

another design issue taken into consideration is the balance between performance and security. Compared with lattice-based PQC, isogeny-based SIKE is relatively slow. A further performance decline that cannot be afforded will prevent its usage, which has to be coped with carefully.

6.1 Security Analysis of SIKE Protected by Existing Countermeasures

Randomized Projective Coordinates. A random number λ can be added to represent the points with different projective coordinates. For example, the projective coordinate $(X : Z)$ without Y can be represented as $(\lambda X : \lambda Z)$, which brings randomization to the point representation. Note that the original point is unchanged with affine coordinates $x = \frac{\lambda X}{\lambda Z}$. Therefore, the coordinates of intermediate values are randomized, and it is effective to resist aforementioned attacks. However, SIKE with only randomized projective coordinate might still be vulnerable to other side-channel attacks such as refined power analysis [41].

Masking the Base Point Q . The input point Q of kernel computation $P + [d]Q$ can be masked by a random point R . Then the isogenies computations can be completed through leveraging the (randomized) tautology $[d](R + Q) - [d]R$. The masking point R is stored in the cryptographic devices and updated by computing $R \leftarrow (-1)^b 2R$ at each execution, where b is a random number generated at each round. Therefore, the adversary can not be aware of the input point $R + Q$, so that the attacks in Section 4 may meet difficulties. However, another extra scalar multiplication $[d]R$ must be calculated, which reduces the speed of SIKE further.

Masking the Scalar d . The scalar d can be masked by a random number r and calculated as $d^* = d + r \cdot \text{Order}(Q)$, where $\text{Order}(Q)$ denotes the order of Q (i.e., the smallest positive integer such that $\text{Order}(Q) \cdot Q = \mathcal{O}$). Therefore we have $[d^*]Q = [d]Q$. So DPA attacks can not succeed due to the randomized scalar. However, it is shown that this method may still leak the scalar under carry-based attack [41].

Random Isomorphic Elliptic Curve. The scalar multiplication can be performed on a random isomorphic elliptic curve first, and then transited back to original curve to conduct subsequent computations. Assuming σ is a random isomorphism from $E[\mathbb{F}_p^2]$ to $E'[\mathbb{F}_{p'}^2]$, the scalar multiplication $[d]Q$ can be computed as $\sigma^{-1}([d](\sigma(Q)))$ [21]. More specifically, the point Q is first mapped to another isomorphic elliptic curve $\sigma(Q)$, and then it is multiplied by the scalar d . Finally, the result is returned to the original curve though σ^{-1} . Even though it is effective to resist aforementioned attacks, the computation of σ and σ^{-1} will take a lot of time and the performance will be reduced accordingly [41].

Random Key Splitting. The scalar d can be formed as $d = [d/r]r + (d \bmod r)$ to bring in the randomization. When computing $P + [d]Q$ in isogeny computation of SIKE, the $P + [d_1]Q + [d_2]Q$ is calculated instead, where $d_1 = d \bmod r$ and $d_2 = [d/r]r$. Similar to the scheme of masking the base point, this method also leads to great performance degradation.

In short summary, there are possible countermeasures which can be adopted from ECC and applied to SIKE. However, when taking both resistance and performance into account, most of them still have certain shortcomings to

some extent. Therefore, it is in demand to propose a new and efficient solution to protect SIKE algorithm against side-channel attacks.

6.2 Our Proposal of Window-Based Countermeasure

In this section, an efficient window-based countermeasure is proposed to eliminate the vertical vulnerability and protect the isogeny computation of SIKE against SCA, with small overhead in time and memory complexity.

In terms of security, two randomized techniques—random initial point and randomized projective coordinates—are both adopted to resist the attacks illustrated in Sections 4 and 5 and other side-channel attacks, such as refined power analysis (RPA). Besides, in order to resist SPA, all the point doublings are performed first, and the results are saved in a table. Then point additions are performed for each non-zero bit. Therefore, power traces of such execution will have a fixed pattern, i.e., a segment of power consumption for point doubling is followed by that for point additions.

In addition to security, we also try to achieve better performance by reducing the time and memory overhead as much as possible. In a scalar multiplication of isogeny computation, the number of point doublings is usually fixed for each bit of the scalar. However, the number of point additions varies in different algorithms and fewer additions lead to better performance, which is determined by the number of non-zero bits in the representation of the scalar.

Hence, in order to reduce the time overhead, the scalar d is represented in a signed binary form as shown in Eq.(7), where $d_i \in \{-1, 0, 1\}$ and m is the number of bits in d .

$$d = \sum_{i=0}^{m-1} d_i 2^i. \quad (7)$$

The point addition is performed when $d_i = 1$. The point subtraction is executed when $d_i = -1$. It should be noted that the performance of point subtraction is similar to point addition. Such subtraction can be executed through a negation followed by an addition, which is relatively fast to compute [42]. More importantly, the Non-Adjacent Form (NAF) is applied to obtain the minimal Hamming weight, which is a kind of binary representation but not allowing two adjacent non-zero digits. In other words, two non-zero bits, 1 or -1 , are separated by at least one zero. How to get the NAF representation of d is presented in Algorithm 2, which is achieved through a recursive function, and the symbol “||” in Line 10 means concatenation of bits. In this way, the total number of point additions/subtractions is the minimal, since the number of non-zero bits is minimized to about one-third of all bits as proved in [43]. Therefore the window-based countermeasure brings the security with minimized time cost.

However this strategy will lead to a large memory overhead because an extra table is introduced to store the results of point doublings. To cope with this problem, the idea of window-based method is proposed [44], in which the bits in binary representation of scalar are processed window by window. In each window, a much smaller value between 0 to $2^w - 1$ is processed when the window size is w . Compared to the window-based method with random initial point

(WBRIP) proposed by Mamiya [44], our method changes the binary representation form of d through NAF algorithm, so that the number of additions is up to the minimum and performance is improved dramatically. In this way, w point doublings are performed first and then the results are saved in a small table, which reduces the memory cost greatly.

Algorithm 2. Function: $NAF()$

Input: Scalar: d

Output: The NAF representation of scalar: d_{NAF}

```

1: if  $d == 0$  then
2:   return ();
3: end
4: if  $d \bmod 2 == 0$  then
5:    $z = 0$ ;
6: end
7: else
8:    $z = 2 - (d \bmod 4)$ ;
9: end
10: return  $NAF((d - z)/2) || (z)$ 
```

Algorithm 3. Window-Based Countermeasure, $w=3$

Input $d_{NAF} = (d_{l-1}, \dots, d_0)$, $P = (x_P, y_P)$, $Q = (x_Q, y_Q)$

Output $\ker(\phi) = \langle P + [d]Q \rangle$

```

1: if  $x_Q == 0$ 
2:    $\ker(\phi) = \langle P + Q \rangle$ ;
3:   return  $\ker(\phi)$ ;
4: end
5: Select a random point  $U$  on the elliptic curve, and  $U \neq Q$ ;
6: Define a temporary variable  $K = (x_K, y_K)$ ;
7:  $K = U$ ;
8: Select  $\lambda$  randomly;
9:  $P = (\lambda X_P : \lambda Z_P) \leftarrow (x_P : 1)$ ;
10:  $Q = (\lambda X_Q : \lambda Z_Q) \leftarrow (x_Q : 1)$ ;
11:  $K = (\lambda X_K : \lambda Z_K) \leftarrow (x_K : 1)$ ;
12: Adding leading 0's in  $d$  to  $m$  bits, so the number of the bits
    in  $d$  is a multiple of 3;
13: for  $i = 0; i < m; i += 3$  do
14:    $T_0 = Q$ ;
15:   for  $j = 1; j \leq 3; j++$  do
16:      $T_j = 2T_{j-1}$ ;
17:   end
18:   switch  $(d_{i+2}, d_{i+1}, d_i)_2$  do
19:     case 0:  $K = K + T_0$ ,  $K = K - T_0$ , break;
20:     case 1:  $K = K + T_1$ ,  $K = K - T_0$ , break;
21:     case 2:  $K = K + T_2$ ,  $K = K - T_1$ , break;
22:     case 3:  $K = K + T_2$ ,  $K = K - T_0$ , break;
23:     case 4:  $K = K + T_3$ ,  $K = K - T_2$ , break;
24:     case 5:  $K = K + T_2$ ,  $K = K + T_0$ , break;
25:     case 6:  $K = K + T_3$ ,  $K = K - T_1$ , break;
26:     case 7:  $K = K + T_3$ ,  $K = K - T_0$ , break;
27:   end
28:    $Q = T_3$ ;
29: end
30:  $K = K - U$ ;
31:  $\ker(\phi) = \langle P + K \rangle$ ;
32: return  $\ker(\phi)$ 
```

After computation, the ratio of the Hamming weight in the window to the window size is always close to 0.6 when w varies from 3 to 9, so the window size w is expected to be as

small as possible to achieve the minimal memory cost. Therefore, a window size of 3 bits is illustrated in Algorithm 3. In order to compute isogeny ϕ in SIKE, its kernel is performed through $\langle P + [d]Q \rangle$. First of all, whether the coordinates of Q are zero or not is checked in Line 1 to 4, so as to resist RPA attacks. Line 5 to 11 show that the random initial point U and randomized projective coordinate with λ are combined to thwart DPA attacks. Here a temporary variable K is defined and initialized as U . If necessary, Line 12 will add leading 0's to d so that the number of bits is a multiple of three ($w = 3$). Then all the bits are processed window by window from Least Significant Bit (LSB) to Most Significant Bit (MSB). Line 15 to 17 mean that three point doublings are first performed to obtain $[2]Q, [4]Q, [8]Q$. The results are stored in a table with four entries— $(T_0, T_1, T_2, T_3) = (Q, [2]Q, [4]Q, [8]Q)$. Which point additions or subtractions to be executed will depend on the value in the current window, as shown in Line 18 to 28. In this way, three doublings followed by two additions/subtractions are always performed for each window if $w = 3$. Afterwards, the random point U is removed from K and $P + [d]Q = P + K$ is obtained. Finally, the kernel of isogeny: $\ker(\phi) = \langle P + K \rangle$ is returned.

6.3 Security Analysis and Performance Evaluation

First of all, the window-based countermeasure performs the same number of doublings and additions/subtractions for each window, as shown in Line 13 to 29 of Algorithm 3. So this countermeasure can resist to SPA and timing attacks effectively.

In addition, our proposed window-based method also demonstrates higher security when facing DPA and RPA attacks. For example, the randomized projective coordinates proposed by Coron fails to resist refined power analysis and zero-value point attacks [41]. Because the zero-point still remains zero, even though the point is represented with randomized projective coordinate. In contrast, the window-based method combines the randomized projective coordinates with the random initial point proposed in [44]. Instead of adding the random point U to the initial point Q , U is directly assigned to a temporary variable K . Since K starts from a random point, the adversary cannot predict the value of K in the process of isogenies computations. To prevent both RPA and ZPA, the special point, which might have a zero coordinate, is always checked. As for masking the scalar, it may still have a lot of bits unchanged due to the fact that the masking $Order(Q)$ has a long run of zeros. So it is still possible to extract a part of bits of the private key [41]. In comparison, our window-based countermeasure exploiting the random initial point and randomized projective coordinates will not be vulnerable to such attacks. Because the original point is blinded with a random point and the zero coordinate will not always appear. Besides, the values in the table (T_0, T_1, T_2, T_3) are also randomized owing to the randomized projective coordinates. For example, T_0 is used to save the base point Q that is randomized in Line 14, so the attacks targeting at those intermediate values in this table will not work.

Furthermore, the SIKE algorithm is relatively slow compared to other PQC algorithms, so the overhead of the countermeasure is a very important metric. For instance, masking

the base point and random key splitting methods cause a great degradation of performance, because another extra scalar multiplication has to be calculated at each round. The computation time is thus doubled. So these two methods are generally not adopted in this paper. In comparison, our proposed method leads to better performance in terms of time. An important property of NAF is that its Hamming weight is minimal among all signed binary representations. On average, 1/2 of the bits in binary representations are non-zero, while only 1/3 of the bits are non-zero in NAF representations [43]. This implies that, in a scalar multiplication, the total number of additions and subtractions is minimal in our countermeasure. Hence, the computation of the isogeny kernel reduces 16.6 percent time cost when window-based strategies are applied.

Moreover, the memory overhead of the window-based countermeasure is a table with $w + 1$ entries, which is linear to w . For example, only four entries are required when $w = 3$. Therefore, the memory cost is very low and reasonable to afford. It is possible to apply this countermeasure to those resource-constrained devices.

In short summary, compared to previous countermeasures, our window-based method can achieve higher security while maintaining a low cost of time and memory. Thus this countermeasure is designed to protect the isogeny computation of SIKE.

7 CONCLUSION

In this paper, we make a thorough security evaluation on the quantum-resistant supersingular isogeny-based cryptosystem in terms of SCA, which is a candidate of Round 2 submission to NIST. First, we introduce a thorough methodology to detect both horizontal and vertical vulnerabilities of SIKE. Then a potential vertical vulnerability in the optimized implementation is analyzed exhaustively in theory, which applies the Three-point Montgomery Differential Ladder algorithm. Subsequently, we propose a specific DPA attack exploiting this vulnerability. Then, practical differential electromagnetic attacks (DEMA) are performed on the 32-bit ARM platform to extract the private key successfully. The experimental results show that the Multi-Bit DEMA can success with only hundreds of electromagnetic samples. And it is also discovered for the first time that the SIKE can be attacked successfully at a sampling rate of Megahertz even if the frequency of core processor reaches to Gigahertz. Finally, for the purpose of eliminating vertical leakages and resisting SCA attacks, an efficient window-based countermeasure is proposed to protect the isogeny computation of SIKE with only a little time and memory overhead. Through detailed security analysis and performance evaluation, our proposed countermeasure leads to higher security and smaller costs compared to most of existing defensive methods.

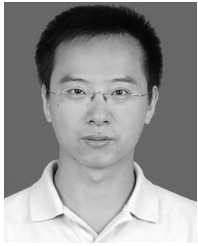
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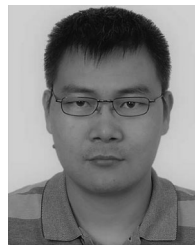
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